

Getting to a mode

Six buttons select a mode directly. Pressing the same button again reaches its hidden sibling where it has one.

	→ Gearbox · press again for XGEAR (cross-feed)
	→ Turning
	→ Facing
	→ Parting
	→ Cone
	→ Threading · press again for TPR (tapered)

M cycles the rest: **A1** (if fitted) → **ELLI** → **GCODE** → **ASY** → **SLOT** → back to the gearbox

The modes

Mode	On screen	Key	What it is for	Stops	Its own settings
Gearbox	—		Electronic lead screw. Carriage feeds from the spindle at the set pitch.	none	—
XGEAR	XGEAR	 ×2	Power cross-feed. Same gearbox, driving the cross slide instead.	none	—
Turning	TURN		Turns to a diameter over several passes, returning between each.	all four	Passes · Spring passes · Clearance
Facing	FACE		Faces the end over several passes. Direction is set in the wizard, not by pitch sign.	all four	Passes · Spring passes · Clearance
Parting	CUT		Parting and grooving: feeds X in, optionally pecking to break the chip.	X only	Passes · Peck depth · Clearance
Threading	THRD		Single-point threading, multi-pass and multi-start. Up limit is full thread depth.	all four	Passes · Spring passes · Flank infeed · Clearance · Thread database
Tapered thread	TPR	 ×2	Threading on a cone — NPT, BSPT. Asks for the taper as an extra wizard step.	all four	as threading, plus its own taper
Cone	CONE		Continuous taper turning at a set ratio. Feed in by hand between passes.	none	taper ratio, asked in the wizard
Ellipse	ELLI		Convex or concave half-spheres and half-ellipses over multiple passes.	all four	Passes
Slotting	SLOT		The lathe as a shaper. Spindle is not involved: Z strokes while X steps down.	all four	Passes · Left reduction
Async	ASY		Feed at a fixed rate, not synchronised to the spindle. Pitch means mm per second.	none	—
G-code	GCODE		Runs a stored program sent from the browser or over USB.	none	—
A1 axis	A1		Jogs and positions the fourth axis. Only in the cycle when A1 is fitted.	none	—

Running an automated mode

1		Set the pitch, then the soft limits. The limits are the cut — for a thread the up limit is full depth.
2		The bottom line reads ON to set up 3 steps . Press play.
3		Answer each step. 1/3 is where you are and how many there are. <> means the arrows change the answer; a number is typed on the numpad. Play accepts.
4		The last step is always Go? , showing how far the tool moves to reach the starting corner. Play starts it; stop returns to the beginning.
5		While running the line reads Pass 2 of 6 with a progress bar. Play skips to the next pass; stop or a jog button ends it.

Thread depth is yours to set: the controller sets the pitch from the database but cannot know how deep. For a 60° thread the depth on the **radius** is $0.6134 \times \text{pitch}$ — M10×1.5 is 0.92mm. In diameter readout mode, enter twice that.

Reading the screen

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THRD* off step 1.00
Pitch 1.50mm x2
Z 12.345 X 25.400
3/3 Go Z-5.00mm?
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- Line 1** Mode, on/off, then a glyph for each soft limit that is set, and the jog step. * means the key has a second mode behind it. During setup it shows the distance between the stops instead.
- Line 2** Pitch, and the number of starts if more than one.
- Line 3** Z and X positions. A slashed X means the readout is diameter, not radius.
- Line 4** Everything else: the setup wizard, pass progress, refusal messages, and whichever readout the display button has selected.

The **display** button cycles what line 4 shows when nothing else needs it: spindle angle → rpm with cutting speed → large two-line position readout → off. Spindle divisions replaces the angle with an index readout; constant speed replaces the cutting speed with the rpm you should be at.

First-time setup

Flash the firmware, then pick metric or inch with the measure button. Open the calibration routines: **hold** the settings button → **Calibration** → play. Work down this list — it runs from zero risk to most, and each routine is more trustworthy once the ones above it are right.

#	Routine	Moves?	What it does
1	Input tester	no movement	Every key reports a code. Confirms the panel before anything moves.
2	Encoder signal	no movement	Coherence should read 100 while the spindle turns. Fix the cable, not the filter.
3	Encoder PPR	you hand-turn	Counts through a known number of turns and derives the pulses per revolution.
4	Encoder direction	no movement	Turn the spindle forward; the count must rise.
5	Z / X direction	axis moves	Confirms which way each axis travels before any measured move.
6	Z / X screw pitch	axis moves	Moves a known distance, you measure it, the pitch is corrected.
7	Z / X backlash	axis moves	Takes up the slack against an indicator to find the lost motion.
8	Z / X travel limit	you jog	Jog to each end to record the usable travel.
9	Z / X max speed	axis moves	Ramps until the motor stalls, then backs off. Do this last.

Then back the settings up — there is no undo. Send \$ over USB to dump every setting as key=value lines, or download the same file from the web page. Paste the lines back to restore.

Global settings

Hold the settings button to open the menu. These belong to the machine; anything belonging to an operation lives on that mode's own page — see page 1.

Preferences		Z axis / X axis / A1 axis	
X readout	Radius or diameter. Diameter doubles every X figure you type and see.	Invert direction	If the axis moves the wrong way.
Retract distance	How far the one-key X retract pulls the tool clear.	Backlash	Lost motion, taken up automatically on a reversal.
Manual step time	How long one tap of a jog key takes.	Lead screw pitch	Travel per screw revolution.
Step rest	Pause between stepped moves.	Motor steps/rev	Including microstepping.
Spindle divisions	Turns the angle readout into an indexer. 0 is off, 6 gives hex flats.	Motor / Lead screw pulley	Teeth either end, for a belt-driven axis.
Index tolerance	How close counts as on the mark, in tenths of a degree.	Start speed	Where the acceleration ramp begins.
Constant speed	The single switch for the cutting-speed readout.	Max speed	Ceiling for rapid and manual moves.
Material	Picks the speed from a table. Manual means use the figure below.	Acceleration	Steps per second squared.
Tool	HSS or carbide. Carbide runs roughly three times as fast.	Max travel	Used for the emergency stop limit.
Manual speed	Surface speed when Material is Manual. Follows metric/inch.	Hold when idle	Off for open-loop drivers that get hot holding position.
Spindle max rpm	Your lathe's ceiling, so an unreachable target is shown capped.	Fitted / Rotary	A1 only: whether it exists, and whether it turns rather than slides.
Spindle encoder		Handwheels	
Encoder PPR	Pulses per revolution as marked. Read four times over.	Fitted 1 / 2	Whether each pulse generator is connected.
Direction	Reverses the count without swapping wires.	Drives 1 / 2	Which axis it turns.
Spindle / Encoder pulley	Teeth either end of the belt, if the encoder is geared.	Invert 1 / 2	If it drives the wrong way.
Divider	Folds several counts into one step to steady a fluttering encoder.	Pulses per rev	Of the handwheel itself.
Dead-band	How far the count must reverse before the axes follow.	Min pulse width	Noise rejection, in microseconds.
Dead-band shape	One-way models screw backlash; symmetric filters both directions.	Dead-band	Stops a reversal being read from a wobble.
Glitch filter	Noise rejection. Too high and real pulses are dropped at speed.	Joystick	
		Fitted	Needs all six auxiliary terminals, so it conflicts with everything else on them.
		Debounce	Contact changes quicker than this are switch bounce.
		WiFi and updates	
		Enabled	Brings up the machine's own access point. Off means no radio at all.
		Access PIN	The WPA2 password, exactly 8 digits. Change it from the default.

From a phone or laptop

- Settings → WiFi and updates → **Enabled** = yes. Change the **Access PIN** from the default first: anyone who can join the network can reflash the machine.
- Join the **NanoEls-H4** network using that 8-digit PIN. The panel shows the address beside the Enabled setting.
- Open **192.168.4.1**. Every setting is editable, grouped as on the panel, with each mode's own settings shown too.
- The strip along the top is live: position, rpm, cutting speed, the constant-speed target, and the encoder **signal** figure. Leave it open through a job and check the low-water mark afterwards.
- Derived figures** shows what your settings actually mean — step resolution, max feed, and the spindle rpm ceiling at the current pitch.
- Firmware updates upload from the same page. Refused unless the machine is stopped; the panel shows progress, not the browser.

Writes are refused while the machine is running, from the panel, USB and the browser alike — stop it first.